



Designation: D4543 – 19

Standard Practices for Preparing Rock Core as Cylindrical Test Specimens and Verifying Conformance to Dimensional and Shape Tolerances¹

This standard is issued under the fixed designation D4543; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

1. Scope*

1.1 These practices specify procedures for preparing rock test specimen of rock core from drill core obtained in the field or from block samples for strength and deformation testing and for determining the conformance of the test specimen dimensions with tolerances established by this practice. Cubical, rectangular, or other shapes are not covered by this practice. However, some of the information contained within this practice and in standard Test Method C170 may still be of use to preparing other test specimen shapes.

1.2 Rock is a complex engineering material that can vary greatly as a function of lithology, stress history, weathering, moisture content and chemistry, and other natural geologic processes. As such, it is not always possible to obtain or prepare rock core specimens that satisfy the desirable tolerances given in this practice. Most commonly, this situation presents itself with weaker, more porous, and poorly cemented rock types and rock types containing significant or weak (or both) structural features. For rock types which are difficult to prepare, all reasonable efforts should be made to prepare a specimen in accordance with this practice and for the intended test procedure. However, when it has been determined by trial and error that this is not possible, prepare the rock specimen to the closest tolerances practicable and consider this to be the best effort (Note 1) and report it as such and if allowable or necessary for the intended test, capping the ends of the specimen as discussed in this practice is permitted.

NOTE 1—Best effort in surface preparation refers to the use of a well-maintained, suitable surface grinder, lathe or lapping machine and any required ancillary equipment are utilized by an experienced operator and in which a reasonable number of attempts has been made to meet the tolerances required in this procedure.

1.3 This practice covers some, but not all of the curatorial issues that should be implemented. For curatorial issues that should be followed before and during specimen preparation

refer to Practices D5079 and to the specific test standards in 2.1 for which the specimens are being prepared.

1.4 This practice also prescribes tolerance checks on the length-to-diameter ratio, straightness of the elements on the cylindrical surface, the flatness of the end bearing surfaces, and the perpendicularity of the end surfaces with the axis of the core.

NOTE 2—This practice does not purport to cover all the issues that will or could be encountered that may control the quality of the specimen preparation required. Each laboratory may have their own issues, especially for different compression load frames or rock types. For example, stiff testing frames versus traditional load frames and loading platens with or without spherical seating. Specimens for a stiff testing load frame with no spherical seat may need to have more stringent requirements depending on the type of rock being tested. This procedure has tried to show the methods and QA that may be involved while keeping in mind those materials that are difficult to work with and for which the specimens will still be suitable to be tested. The available literature and input on this subject from D18.12 members were considered as much as possible for this standard.²

1.5 The requirement for specifying the moisture condition and volume of the test specimen is also stated. However, the requirements in the specific test standards in 2.1 should be followed too.

1.6 All observed and calculated values shall conform to the guidelines for significant digits and rounding established in Practice D6026, unless superseded by this standard.

1.6.1 The practices/procedures used to specify how data are collected/recorded and calculated in this standard are regarded as the industry standard. In addition, they are representative of the significant digits that generally should be retained. The procedures used do not consider material variation, purpose for obtaining the data, special purpose studies, or any considerations for the user's objectives; and it is common practice to increase or reduce significant digits of reported data to be commensurate with these considerations. It is beyond the scope

¹ These practices are under the jurisdiction of ASTM Committee D18 on Soil and Rock and are the direct responsibility of Subcommittee D18.12 on Rock Mechanics.

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² Needless Stringency in Sample Preparation Standards for Laboratory Testing of Weak Rocks, P.J.N. Pells (Coffey & Partners Pty Ltd, North Ryde) | M.J. Ferry (Postgraduate Scholar, University of Sydney), International Society for Rock Mechanics Source 5th ISRM Congress, 10-15 April, Melbourne, Australia Publication Date 1983.

*A Summary of Changes section appears at the end of this standard

of this standard to consider significant digits used in analysis methods for engineering design.

1.7 Units—The values stated in inch-pound units are to be regarded as standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard. Add if appropriate, “Reporting of test results in units other than inch-pound shall not be regarded as nonconformance with this standard.”

1.7.1 The slug unit of mass is typically not used in commercial practice; that is, density, balances, and so on. Therefore, the standard unit for mass in this standard is either kilogram (kg) or gram (g) or both. Also, the equivalent inch-pound unit (slug) is not given/presented in parentheses.

1.7.2 It is common practice in the engineering/construction profession to concurrently use pounds to represent both a unit of mass (lbm) and of force (lbf). This practice implicitly combines two separate systems of units; the absolute and the gravitational systems. It is scientifically undesirable to combine the use of two separate sets of inch-pound units within a single standard. As stated, this standard includes the gravitational system of inch-pound units and does not use/present the slug unit for mass. However, the use of balances or scales recording pounds of mass (lbm) or recording density in lbm/ft³ shall not be regarded as nonconformance with this standard.

1.8 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.9 *These practices offer a set of instructions for performing one or more specific operations. This document cannot replace education or experience and should be used in conjunction with professional judgement. Not all aspects of this practice may be applicable in all circumstances. This ASTM standard is not intended to represent or replace the standard of care by which the adequacy of a given professional service must be judged, nor should this document be applied without consideration of a project’s many unique aspects. The word “standard” in the title of this document means only that the document has been approved through the ASTM consensus process.*

1.10 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 ASTM Standards:³

C170 Test Method for Compressive Strength of Dimension Stone

C617 Practice for Capping Cylindrical Concrete Specimens
D653 Terminology Relating to Soil, Rock, and Contained Fluids

D2113 Practice for Rock Core Drilling and Sampling of Rock for Site Exploration

D2216 Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass

D2936 Test Method for Direct Tensile Strength of Intact Rock Core Specimens (Withdrawn 2017)⁴

D3740 Practice for Minimum Requirements for Agencies Engaged in Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction

D4753 Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing

D5079 Practices for Preserving and Transporting Rock Core Samples (Withdrawn 2017)⁴

D6026 Practice for Using Significant Digits in Geotechnical Data

D7012 Test Methods for Compressive Strength and Elastic Moduli of Intact Rock Core Specimens under Varying States of Stress and Temperatures

D7070 Test Methods for Creep of Rock Core Under Constant Stress and Temperature

3. Terminology

3.1 Definitions:

3.1.1 For definitions of common technical terms used in this standard, refer to Terminology **D653**

4. Significance and Use

4.1 The dimensional, shape, and surface tolerances of rock core test specimens are important for determining rock properties of intact specimens. This is especially true for strong rocks, greater than 7250 psi (50 MPa) and for rock specimens that will be tested in stiff testing load frames without a spherical seat where non-uniform loading could occur. Dimensional and surface tolerance checks are required in the test methods listed in Section 2.1. To simplify test procedures in laboratories, the parts of those procedures that are common to the test methods in Section 2.1 are given in this standard.

4.2 This procedure is applicable to all the standards listed in Section 2.1; however, specimens for Test Method **D2936** do not need to be machined or to meet the specified tolerances for flatness and parallelism.

4.3 The moisture condition of the specimen at the time of the sample preparation can have a significant effect upon the strength and deformation characteristics of the rock. Good practice generally dictates that laboratory tests be made upon a specimens’ representative of field conditions. Thus, it follows that the field moisture condition of the specimen should be preserved until the time of the test. In some instances, however, there may be reasons for testing specimens at other moisture

³ For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard’s Document Summary page on the ASTM website.

⁴ The last approved version of this historical standard is referenced on www.astm.org.

contents, from saturation to dry. In any case, the moisture content of the test specimen should be tailored to the problem at hand.

NOTE 3—Discussions on moisture content are common in many rock testing standards but professional judgement will be needed to both handle and report this issue. For example, when obtaining the samples or preparing the specimens, water or some other cooling agent may be required or used. Therefore, the moisture in the specimen or samples may not be what it was in situ; this applies to both water chemistry and quantity of fluids. This issue should be addressed, and a plan put in place for each step from the sampling to the testing phase in a manner that records/reports what steps were advised to successfully prepare testable samples. Usually a compromise between preserving in-situ conditions, costs, conditions outside the control of the laboratory and obtaining testable specimens is required. For example, loss of moisture that leads to the samples or specimens falling apart may be of greater concern than testing with in situ water or at the in situ water content or both.

4.4 Excess moisture will affect the adhesion of resistance strain gages, if used, and the accuracy of their performance. Adhesives used to bond the rock to steel end caps and fixtures for attaching specimens to actuators and crosshead of the load frame in the direct tension test (D2936) will also be affected adversely by excess moisture.

NOTE 4—The quality of the result produced by these practices is dependent upon the competence of the personnel performing it and the suitability of the equipment and facilities used. Agencies that meet the criteria of Practice D3740 are generally considered capable of competent and objective testing and sampling. Users of these practices are cautioned that compliance with Practice D3740 does not in itself assure reliable results. Reliable results depend on many factors; Practice D3740 provides a means of evaluating some of those factors.

5. Apparatus

5.1 *Support Surface*—A flat test surface which shall not depart from a plane by more than 0.0005 in. (0.013 mm) or meets ISO 9001 Certification. It shall have a large enough area such that the cylindrical surface of a rock core test specimen may be rolled and a V-block end of a rock core test specimen, or displacement gage assembly can be placed (Figs. 1 and 2) to perform the required tolerances measurements. Machinist

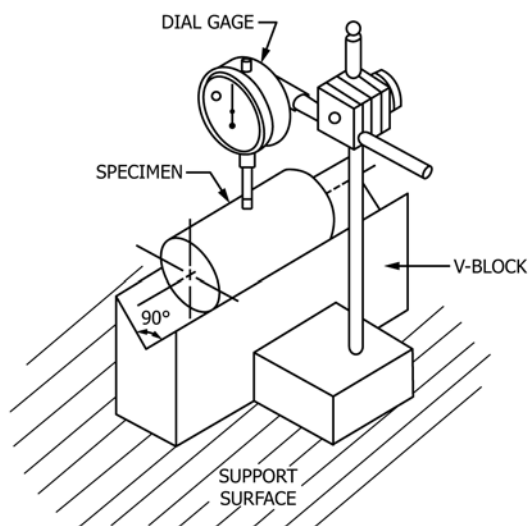


FIG. 1 Basic Dial Gage Setup for Determining the Straightness of Elements Along the Cylindrical Surface (S2)

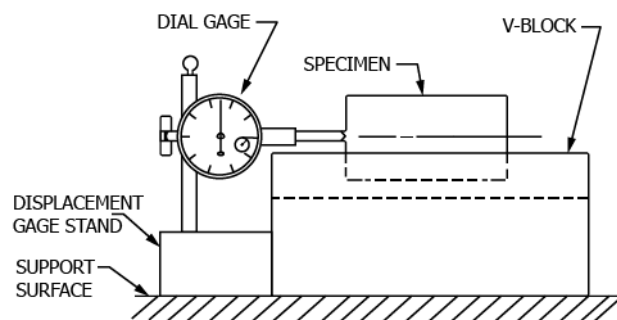


FIG. 2 Assembly for Determining the Flatness and Perpendicularity of End Surfaces to the Specimen Axis (FP1)

grade, certified, granite blocks are commonly used for support surfaces because they do not dent or rust. However, other materials may be used if they meet the criteria of the procedure. The area of the support surface will depend on the size of specimen to be prepared; however, a 12 in. × 12 in. (300 mm × 300 mm) area will be sufficient for most applications.

5.2 *V-block (conformance tests)*—The V-block (Figs. 1 and 3) shall be machinist quality with all bearing faces surfaces ground flat, smooth to within 0.0005 in. (13 μm) and with a 90° included angle. The V-block shall have some means of securing the specimen firmly in the V-block. The dimensions of the V-block must be such that it does not physically interfere with the displacement gage readings and suited for the size of specimen to be handled.

5.3 *Displacement Gage Assembly (Figs. 1 and 2):*

5.3.1 *Dial or Electronic (Contact or Non-contact) Displacement Gage*—The sensitivity of the displacement gage shall be at least 0.001 in. (0.02 mm) for measurement of cylindrical surfaces. The measurement contact tip of the displacement gage should be round. A displacement gage readable to 0.0001 in. (0.002 mm) is advised for measurements on the end surfaces.

5.3.2 *Dial or Electronic (Contact or Optic) Displacement Gage Stand*—A stand with a base and vertically mounted rod with an adjustable gage holder to support the gage on the flat surface at the proper height for the specimen and to take measurements normal to the flat surface. The side of the base can be machined flat so that it may be used as a straight edge for taking measurements as shown in Fig. 1 and Fig. 2 and described in 9.1 and 9.2. See Note 11 for more useful information.

5.4 *Feeler Gage Set*—25 or 26 leaf/blade set; 3 in. (76 mm) long by ½ in. (13 mm) wide, and thicknesses beginning at 0.0015 in. (0.04 mm) and ending at 0.025 in. (0.64 mm).

5.5 *Surface Grinder*—Any manual or automatic machinist's grade surface grinder equipped with a grinding wheel suited for the type and size of specimen, a magnetic flat surface and a V-block (5.6) to hold one or more specimens during the grinding process is suitable. The apparatus is also equipped to apply appropriate cooling and cutting agents (if needed) at the cutting surface to cool the grinding wheel surface and remove any cuttings.

NOTE 5—A commonly available apparatus and method is presented in



FIG. 3 One example of Holding Jig for Surface Grinder that can be Rotated and Grind Both Ends of up to Four Specimens (The adjacent short core is used as a spacer to fill unused specimen slot and yellow on top of specimen is placed prior to grinding for showing where grinding is occurring as the surface grinding progresses. Once lapping of the visible specimen end is completed the top half of the jig is unbolted and the section rotated so the opposing end of the specimen is now facing up and can be lapped as well.)

the sections provided here. Any other specialty or expensive type equipment or methods or combination thereof that can shape the rock core specimen into a specimen and meet the specification is acceptable.

NOTE 6—Surface grinders with greater mass is preferred because they have a greater resistance against vibration and will therefore produce a more uniform surface. Grinding wheels should be true and balanced. Some surface grinders have wheel hubs/flanges that did not have balancing systems however this could be added later using aluminum leaves.

5.6 V-Block, or Other Similar Holding Device (Grinder Accessory)—A metal V-block of sufficient size for holding the rock specimen in the surface grinder, with or without a magnetic chuck and is configured so that the specimen can be rotated to grind both ends and without losing any alignment with the grinder tolerances that would affect the grinding results on the specimen. See showing one example of V-block designed for up to four specimens on a magnetic chuck.

5.7 Lapidary/Trim Saw—A power saw of adequate size and horse power equipped with a circular diamond saw blade

suitable for the material to be cut, with a movable platform for holding and feeding the core specimen, perpendicular to the core axis and feeding the specimen into the cutting surface of the blade. The movable platform may be a manual or automatic feed device. A GFI (Ground Fault Interrupt) for electrical powered saws is advised. The apparatus should be equipped to safely apply appropriate cooling agents (if needed) to the cutter to cool the blade and remove any cuttings and, if required, retained in a settling tank setup for disposal.

5.8 For Drilling Block Samples:

5.8.1 At least a 10-horsepower (750 kgf-m per second) drill, with a GFI (Ground Fault Interrupt) for electrical powered drills is recommended.

5.8.2 Drill can be hand held but some type of rigid platform is preferred, and side straightness may not be met and could then involve extra effort such as machining of sides of the sample or specimen.

5.8.3 Drill bit can be advanced manually or automatically.

5.8.4 A thin walled diamond core barrel is typically used. However, if needed, most drilling supply companies can build a short, double or triple tube core barrel that may be useful for certain rock types too.

5.8.5 A swivel and adaptors for hooking up to the drill to supply a fluid to cool the cutting surface and remove any cuttings.

5.9 (Optional) *Lapper*.

5.10 (Optional) *Machinist Shaper*.

5.11 *Cooling or Cutting Agents*—This is typically portable water or house air but may be some other liquid or gas for special circumstances. Whatever cooling or cutting agent is used it must be suitable for the specimen preparation process such that it leaves a specimen's strength properties unaffected. For example, sedimentary rocks originally deposited in a saline environment may be affected if subjected to fresh water.

5.12 *Machinist Calipers*, or similar device, with vernier, digital, or dial readouts readable to 0.001 in. (0.025 mm) and large enough for the size of the specimens being measured.

5.13 *Scale or Balance*, of adequate size and capacity to measure the mass of the specimen after preparation is completed to the nearest 1 gram. All balances must meet the requirements of Guide **D4753** and this section. A Class GP1 balance of 0.01 g readability is required for specimens having a mass of up to 200 g (excluding mass of specimen container) and a Class GP2 balance of 0.1 g readability is required for specimens having a mass over 200 g. If desired, a Class GP1 balance may be used for specimens exceeding 200 g providing the specimen size is within the capacity of the balance. However, the balance used may be controlled by the number of significant digits needed (see 1.10).

5.14 *Miscellaneous Tools*, machinist scribe, rubber mallet, safety glasses, hearing protection, containers or plastic bags for curatorial control and indelible markers.

6. Samples

6.1 Samples for preparing specimens can be either drill cores obtained directly from the in situ rock (**D2113**) or obtained from block samples cored in the field or in the laboratory.

6.2 Samples should be selected or obtained (or both) to meet the objectives of the specific standard listed in **2.1** and the test program and any requirements related to anisotropic properties, moisture, and alike, of the in situ material that are relevant to the intended use.

6.3 Block samples must be provided which are clearly marked and if needed have appropriate measurements taken during sample acquisition to identify the orientation and or location of any geologic characteristics that might be important for coring samples from the block sample.

7. Specimens

7.1 Test specimens shall be right circular cylinders within the tolerances specified herein.

7.2 The specimen shall have a length-to-diameter ratio (L/D) of 2.0 to 2.5 and a minimum diameter 1-7/8 in. (47 mm) or as directed by the client.

NOTE 7—The larger the internal friction angle of a specimen the more important it will be to have larger L/D ratios so that the specimen can potentially develop a true shear plane that does not pass through either end of the specimen or is not altered by the specimen size.

NOTE 8—It is important to determine what type of test specimen and test apparatus will be used before starting the specimen preparation process or the required end product may not be obtained.

NOTE 9—It is desirable that the diameter of rock test specimens be at least ten times the diameter of the largest discrete mineral grain, particle or crystalline. For weak rock types which behave more like soil (for example, poorly cemented sandstone), the specimen diameter should be at least six times the maximum particle diameter. It is considered that the specified minimum specimen diameter of approximately 1-7/8 in. (47 mm) will satisfy this criterion in the majority of cases. When cores with smaller diameters than the specified minimum must be tested because larger diameter cores are unavailable or impractical use of large drilling equipment (as is often the case in the mining industry), costs, or both, are prohibitive, a suitable notation of this fact shall be made in the report.

7.3 The cylindrical surfaces of the specimen shall be generally smooth and free of abrupt irregularities that are not part of the rock natural fabric, that is, voids or vugs, with all the elements straight to within 0.020 in. (0.50 mm) over the full length of the specimen, as determined by **9.1**, procedure S1 or S2.

7.4 The ends of the specimen shall be cut parallel to each other and at right angles to the longitudinal axis. The end surfaces shall be surface ground or lapped flat to a tolerance not to exceed 0.001 in. (25 µm), as determined by **9.2** and to make the two ends more parallel to each other.⁵

7.5 The use of capping materials or end surface treatments other than the grinding, lathe, and lapping specified herein is not permitted, except as noted in **7.6**.

7.6 There are some rock types with physical characteristics or low strengths which preclude preparing specimens to the flatness tolerance specified in **7.4**, even with the best effort (**Note 9**). In these instances, first cut the core specimen to length, then apply end caps to the end surfaces of the specimen.

7.7 The specifications for the capping compound, capping plates, and alignment devices and the procedure for capping weak rock core specimens shall be the same as those established for compression testing of concrete in Practice **C617**; however, melted sulfur capping compounds are not permitted because of the possible detrimental effects of the high temperature from such capping compounds on the rock. Dental plaster and high strength gypsum cements are commonly used but other materials may be acceptable or better suited for the rock types being prepared.

7.8 The ends of the specimen shall not depart from perpendicularity to the axis of the specimen by more than 0.25° as determined in Section **9**.

7.9 The parallelism tolerance is the maximum angular difference between the opposing best-fit straight line on each

⁵ Hoskins, J. R., and Horino, F. G., "Effects of End Conditions on Determining Compressive Strength of Rock Samples," *Report of Investigations U.S. Bureau of Mines 7171*, 1968.

specimen end. It shall not be more than 0.25° for spherically seated test machines and 0.13° for fixed end test machines as determined in Section 9.⁵

8. Procedure for Preparing Test Specimens

8.1 Specimens can be prepared from either drill cores obtained directly from the in situ rock or obtained from block samples cored in the field or in the laboratory. If block samples are being used, then Section 8.2 which describes laboratory coring of specimens from block samples should be followed before proceeding to Section 8.3. Practice D2113 describes rock core drilling and sampling of rock for site investigations. Water is normally a suitable fluid for rock coring, cutting and grinding operations. However, some rock materials are sensitive to water, the chemistry of the water, or both. For example, using fresh water on saline sedimentary rocks may cause the sample or specimen to fall apart. Therefore, alternate suitable cooling and flushing fluids should be used. In Sections 5.8 and 5.9 an air-cooled grinding unit with a dust collector is recommended for weak rocks and rocks that may react to fluids.

8.2 *Core Drilling Block Samples*—Use a drill as recommended in 5.7 to obtain rock core samples from the block sample(s). In general, surface set diamond thin-wall bits are suited for soft rock and impregnated diamond thin-wall bits are better suited for hard rock. However, experience and trial and error may be required to obtain suitable samples.

8.2.1 Install the core bit and water swivel into the drill press chuck. Give the end of drill bit a tap with a rubber mallet to make sure it is snug.

8.2.2 Lower the core bit to the drill table and mark the core barrel for reference for sufficient drilling depth.

8.2.3 Connect the cooling fluid hose to the swivel and tie it out of the way.

8.2.4 Place a sheet of ½ in. (12.7 mm) or thicker plywood on the drill table. Place and orient if required, the rock block on the plywood and then clamp the rock block securely to the table with clamping devices such as chain vise locks. Block with wood wedges as necessary to make sure the rock is secure and has a relatively flat surface for drill bit to start cutting on.

8.2.5 Turn on the cooling fluid with sufficient flow to cool the bit and to flush the cuttings.

8.2.6 With the bit raised off the sample, turn on the drill using a slow speed.

8.2.7 Lower the bit slowly onto the sample using a slow rotation speed until a groove is started. Use enough down force to prevent chatter but do not allow electric motors to slow so much as to buzz, which could lead to damaging the motor. A loss of drill cooling fluid and reaching the reference position mark indicates the end of the run.

8.2.8 After breaking through, back the bit out of the hole and turn off the drill. If the core is not completely drilled through, remove the block and tap the bottom gently, then remove the core.

8.2.9 Properly document the core sample and store the core.

8.3 Specimen Cutting:

8.3.1 Automatic feed diamond saws are recommended for cutting rock specimens.

8.3.2 Clamp the core sample in the jig used for holding or feeding the specimen during the cutting process for each end of the core sample.

8.3.3 Apply cooling fluid continuously to cool the blade and flush cuttings during the cutting process.

8.3.4 Turn on the saw and use either the manual or automatic feed capabilities of the saw to cut the specimen perpendicular to its axis (see 7.2 and 7.4) at a rate that avoids blade chatter.

8.3.5 Once the specimen is cut, back off the blade and turn off the saw.

8.3.6 Unclamp the specimen and if needed reposition the sample so that the opposite end of the intended specimen can be cut.

8.3.7 Repeat steps 8.3.2 through 8.3.6.

8.3.8 Remove, label and store the specimen after this first stage of specimen preparation according to the test program requirements.

NOTE 10—When cutting weak, soluble, or friable rock such as potash, shale, for example, it is recommended the core be protected as best as possible before cutting to protect the sample from damage from fluids or vibrations. Heat shrink tubing has been successfully used by even the International Ocean Discovery Program (IODP) for marine sediments but the user needs to determine prior to use if the heat will affect the specimen in any deleterious manner.

8.4 *Cylindrical Surface Grinding*—The quality of the circumferential surfaces of core specimens is usually acceptable for most rock types, and no further surface finishing is required. If the circumferential surface contains abrupt or wavy irregularities however, further finishing is recommended. This can be accomplished by surface grinding in a lathe in much the same way as dog bone specimens are prepared for direct tensile tests (D2936). The lathe chuck and center spindle are fitted with brass centers having knurled end bearing surfaces. The specimen is held between the brass centers by end pressure. A tool post grinder equipped with a diamond impregnated wheel is used to grind the cylindrical surfaces. Diamond impregnated grinding wheels are best suited for grinding rock surfaces. Use cooling fluid to cool the surfaces and flush away cuttings.

8.5 *End Surface Flatness and Parallelism*—There are several ways to prepare the end surfaces of a specimen.

8.5.1 Method ES1—Surface Grinding:

8.5.1.1 T core specimen is clamped in a V-block, or similar holding jig, suitable for placement on a machinist table grinder. The V-block is held in place by a magnetic table or other suitable means of holding the V-block under the cutting wheel.

8.5.1.2 The specimen ends are typically colored with a waterproof marker, prior to the start of grinding, in order to monitor where material is being removed.

8.5.1.3 Grind each end one at a time with a diamond impregnated grinding wheel. Silicon carbide grinding wheels or other grinding wheels may also be used if suitable for the type of specimens being prepared.

8.5.1.4 Use a cooling fluid to cool the specimen and grinding wheel surfaces and flush away the cuttings.

8.5.1.5 Grinding increments of 0.002 in. (0.50 mm) is recommended unless experience shows otherwise.

8.5.1.6 Grinding is completed as evidenced by grinding striations covering the whole end surface and where the waterproof coloring is completely removed.

8.5.1.7 Reverse the core and grind the other end. Alternatively, the core may be secured in the chuck of a lathe, and the end surfaces finished using a tool post diamond grinding wheel.

8.5.2 *Method ES2 Lapping*—Lapping devices are available that lap both ends of the core simultaneously using slurried grit.

8.5.3 *Method ES3 Machinist Shaper*:

8.5.3.1 For large specimens, a 4 to 12 in. (100 to 300 mm) in diameter machinist shaper may be used.

8.5.3.2 The shaper is retrofitted with a segmented diamond saw blade mounted on a tool post grinder.

NOTE 11—While these other methods are suitable for some purposes and may be the only method for large diameter specimens, experience has shown that lapping devices do not usually get the results required for this standard. Also, for specimens which are susceptible to the effects of water the slurry grit may also be an issue.

9. Procedure for Verifying Shape Conformance

9.1 *Side Straightness*—Determine the deviation from straightness of the elements by either Procedure S1 or Procedure S2, as follows:

9.1.1 *Procedure S1*:

9.1.1.1 Lay the cylindrical specimen on its side, on a smooth, flat surface.

9.1.1.2 Roll the cylindrical specimen and measure the height of the maximum gap between the specimen and the flat surface with a 0.020 in. (0.50 mm) feeler gage.

9.1.1.3 If the maximum gap exceeds 0.020 in. (0.50 mm), the specimen does not meet the required tolerance for straightness of the elements.

9.1.2 *Procedure S2*:

9.1.2.1 Place the cylindrical surface of the specimen on a V-block that is laid on a flat surface as shown in Fig. 1. The length of the V-block shall be sufficient that the specimen will not project over its ends during movement.

9.1.2.2 Place a dial gage in contact with the top of the specimen cylindrical surface, as shown in Fig. 1, and observe the dial reading as the specimen is moved from one end of the V-block to the other along a straight line, without rotation. Record the maximum and minimum readings of the dial gage and calculate the difference, Δ_0 . If the dial gage traverses a natural cavity in the rock, readings in that region shall not be included in the determination of Δ_0 .

9.1.2.3 Repeat the same operations by rotating the specimen for every $120 \pm 1^\circ$, and obtain the differences Δ_{120} and Δ_{240} . The maximum value of these three differences shall be less than 0.020 in. (0.50 mm).

9.2 Check the end flatness and parallelism tolerance by either Procedure FP1 or Procedure FP2, as follows:

9.2.1 *Procedure FP1*:

9.2.1.1 Place the specimen in a V-block with the dial gage mounted at the end as shown in Fig. 2. Move the dial gage stand or V-block horizontally so that the dial gage measurement tip runs across a diameter of the specimen end surface. Make sure that the dial gage base maintains intimate contact with the end surface of the V-block during movement or any other guide used to control the horizontal movement. Record the dial gage readings across the diameter at least 10 equally spaced locations, except where voids are located, the spacing for that reading should be adjusted to avoid any voids. These readings may be recorded in tabular form, or to simplify the procedure, they may be plotted directly on a graph as shown for End 1, Diameter 1, in Fig. 4. Data recording is simplified if the dial gage is set to zero when it is in contact with the center of the end face.

NOTE 12—A digital displacement gage that is manually read or capable of communicating with a data acquisition system can be used. Also, the tip of the dial gage can also be a roller type tip as well. Ball point type tips have been problematic and are not recommended. Also, a spindle lifting knob or cable option is useful for lifting the tip prior to the movement of the specimen under the displacement gage to eliminate wear on the tip and avoid letting the tip fall into a vug or void and damaging the gage spindle.

9.2.1.2 Plot the readings and draw a smooth curve through the points to represent the surface profile along the specified diametrical plane, as shown for End 1, Diameter 1, in Fig. 4. Do not plot dial gage readings taken when the gage tip drops into a natural cavity in the rock. Rotate the specimen $90 \pm 1^\circ$ about its longitudinal axis and repeat the same operations and tolerance checks for the new diametrical plane.

9.2.1.3 Turn the specimen end for end and repeat the same measurement procedures and tolerance checks for the other end surface.

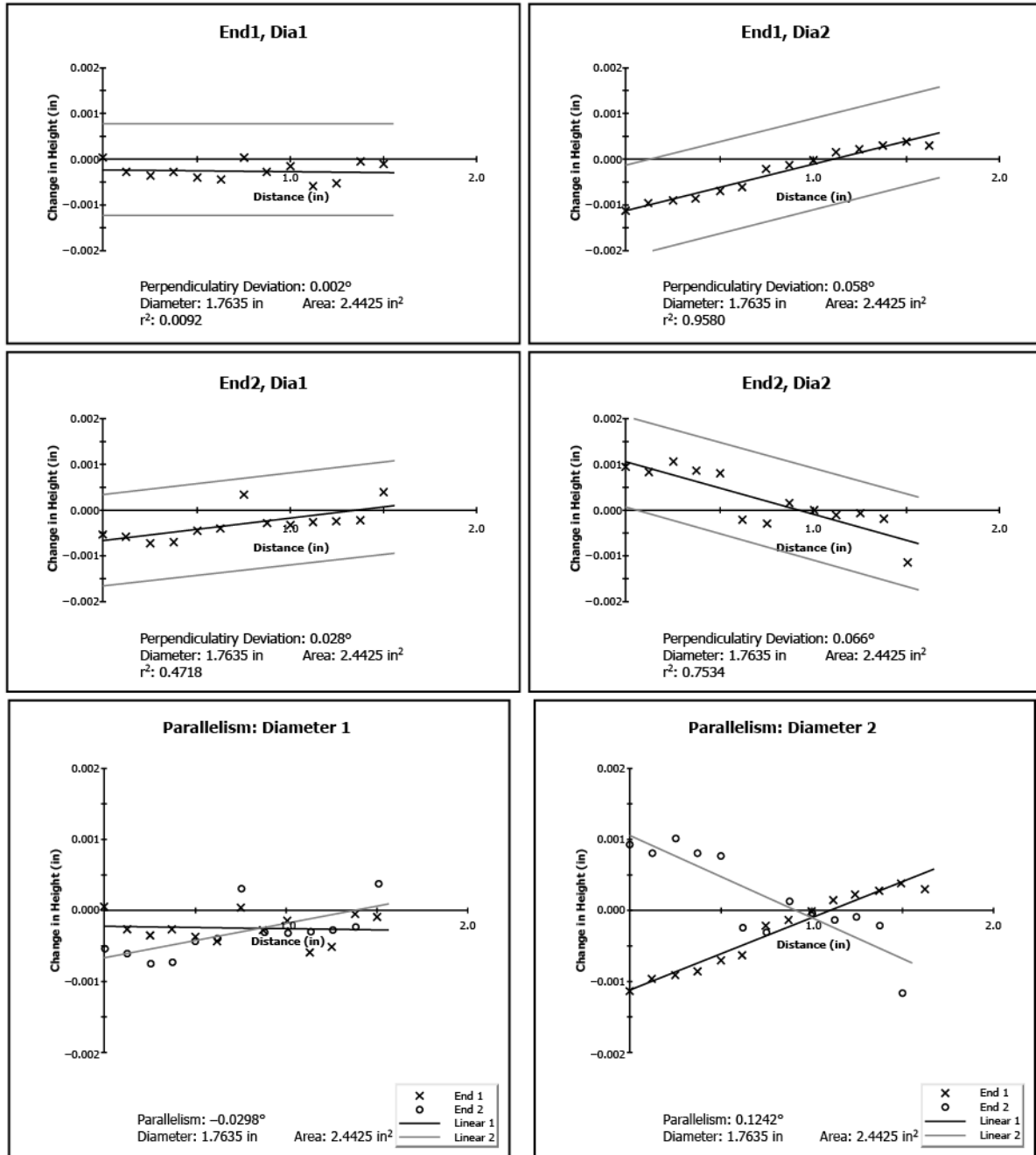
9.2.1.4 The flatness tolerance is met when each smooth curve so determined does not depart from a visual best-fit straight line by more than 0.001 in. (25 μ m). The parallelism tolerance is met when the maximum angular difference between the opposing best-fit straight line on each specimen end is not more than 0.25° for spherically seated test machines and 0.13° for fixed end test machines. These measurements are also used to check perpendicularity in 9.3.1.

Project: Big Valley

Feature: Proposed Big Valley
Dam - RN-274

Drill Hole: 06-2

Depth: 96.9 ft



(a)

FIG. 4 Suggested Format for Graphing End Flatness, Parallelism, Perpendicularity (a), Tables (b), and Summary Table (c) for Presenting Shape and Dimensional Tolerance Data; Methods S1, FP1 and P1 are shown in this example.

Project: Big Valley

Feature: Proposed Big Valley Dam - RN-274

Drill Hole: 06-2

Depth: 96.9 ft

End1_Dia1 - Travel (in)	End1_Dia1 - Displacement (in)	End1_Dia2 - Travel (in)	End1_Dia2 - Displacement (in)	End2_Dia1 - Travel (in)	End2_Dia1 - Displacement (in)	End2_Dia2 - Travel (in)	End2_Dia2 - Displacement (in)
0.000	0.0000	0.000	-0.0011	0.000	-0.0005	0.000	0.0009
0.125	-0.0003	0.125	-0.0010	0.125	-0.0006	0.125	0.0008
0.250	-0.0004	0.250	-0.0009	0.250	-0.0007	0.250	0.0010
0.375	-0.0003	0.375	-0.0009	0.375	-0.0007	0.375	0.0008
0.500	-0.0004	0.500	-0.0007	0.500	-0.0004	0.500	0.0008
0.625	-0.0004	0.625	-0.0006	0.625	-0.0004	0.625	-0.0002
0.750	0.0000	0.750	-0.0002	0.750	0.0003	0.750	-0.0003
0.875	-0.0003	0.875	-0.0001	0.875	-0.0003	0.875	0.0001
1.000	-0.0001	1.000	0.0000	1.000	-0.0003	1.000	0.0000
1.125	-0.0006	1.125	0.0001	1.125	-0.0003	1.125	-0.0001
1.250	-0.0005	1.250	0.0002	1.250	-0.0003	1.250	-0.0001
1.375	-0.0001	1.375	0.0003	1.375	-0.0002	1.375	-0.0002
1.500	-0.0001	1.500	0.0004	1.500	0.0004	1.500	-0.0012
		1.625	0.0003				

(b)

FIG. 4 Graphing End Flatness, Parallelism, Perpendicularity (a), Tables (b), and Summary Table (c) for Presenting Shape and Dimensional Tolerance Data; Methods S1, FP1 and P1 are shown in this example. (continued)

SUMMARY OF COMPRESSION SPECIMEN TOLERANCE TESTING

Background

Method used: ASTM D4543-08
 Project: Big Valley
 Feature: Proposed Big Valley Dam - RN-274
 Checked by: JAT Date: 5/10/2008
 Test Data saved as: C:\Rock Surface\Tests\Proposed Big Valley Dam - RM-274, DH-06-2, 96.9 (4-12-2008 1'48'44 PM).rsl

Specimen Information

Index Number: 67U-19
 Drill Hole: 06-2
 Depth: 96.9 ft
 Mass: 453 g
 Bulk S.G.*: 2.67
 Length: 4.2486 in
 Diameter: 1.7635 in
 Area: 2.44 in²
 L/D Ratio: 2.41
 Tested by: S. Reo
 Date Tested: 4/12/2008

Test Results

<u>Parameter</u>	<u>Data</u>	<u>Pass/Fail</u>
Side Tolerance (Straightness):	0.0015 in	Passed
Perpendicularity Deviation		
End 1, Diameter 1:	-0.0023°	Passed
End 1, Diameter 2:	0.0580°	Passed
End 2, Diameter 1:	0.0276°	Passed
End 2, Diameter 2:	-0.0662°	Passed
Max. Deviation from flatness:	0.0006 in	Passed
Parallelism Deviation		
Diameter 1:	-0.0298°	Passed
Diameter 2:	0.1242°	Passed

Tolerance Limits

Side Tolerance (Straightness): Not to exceed 0.020 in
 Perpendicularity Deviation: Not to exceed 0.250°
 Deviation from Flatness: Not to exceed 0.001 in
 Parallelism Deviation: 0.13° or less for fixed spherical seat

Notes

* Specific Gravity is calculated using a pure water density at 20°C

(c)

FIG. 4 Graphing End Flatness, Parallelism, Perpendicularity (a), Tables (b), and Summary Table (c) for Presenting Shape and Dimensional Tolerance Data; Methods S1, FP1 and P1 are shown in this example. (continued)

9.2.2 Procedure FP2:

9.2.2.1 Set the specimen upright on a flat surface.

9.2.2.2 Do a rapid check on the specimen to see if it will pass flatness tolerances by placing a dial gage measurement tip in contact with the top of the specimen. Move the dial gage measurement tip across the top of the specimen along at least three different diameters. Note the maximum and minimum dial gage readings. Do not include dial gage readings taken when the measurement tip drops into a natural cavity in the rock.

9.2.2.3 Turn the specimen end-for-end and repeat.

9.2.2.4 The flatness tolerance is met when the smooth curve so determined does not depart from a visual best-fit straight line by more than 0.001 in. (25 µm).

9.2.2.5 From the rapid flatness tolerance check decide whether to proceed with checking the specimen tolerance or to have the specimen reground or go to the next step.

9.2.2.6 Scribe two lines, 90 degrees to each other on both ends of the specimen for taking measurements. The lines on each end of the specimen should be mirror images of each other. If a diametrical high and low spot exist, the lines should be selected to capture the worst-case tolerance.

9.2.2.7 Set the specimen upright on a flat surface and using a displacement gage assembly record at least 10 evenly spaced dial gage readings across each of the two diameters.

9.2.2.8 Turn the specimen end-for-end and repeat.

9.2.2.9 The parallelism tolerance is met when the maximum angular difference between the opposing best-fit straight line on each specimen end is not more than 0.25° for spherically seated test machines and 0.13° for fixed end test machines.

9.3 The ends of the specimen shall not depart from perpendicularity to the axis of the specimen by more than 0.25°, which is a slope of 1 part in 230.^{5,6} Check this tolerance using either Procedure P1 or Procedure P2, as follows:

9.3.1 Procedure P1—Use the measurements taken in 9.2.1. Calculate the difference between the maximum and the minimum readings on the dial gage along Diameter 1. This difference is denoted as Δ_1 . Calculate the corresponding difference for Diameter 2, which is 90° from Diameter 1. Denote the difference for Diameter 2 as Δ_2 . Calculate the corresponding differences for the other end of the specimen, Δ_1' and Δ_2' . The perpendicularity tolerance will be considered to have been met when:

$$\frac{\Delta_i}{d} \text{ and } \frac{\Delta_i'}{d} \leq \frac{1}{230} = 0.0043 \quad (1)$$

where:

i = 1 or 2,

' = opposite flat end of core specimen,

d = diameter of specimen (see Section 10.2), and

Δ_i = difference in displacement measurements normal to the specimen end surface.

9.3.2 Procedure P2—Set the specimen upright on a smooth flat surface. Place the base of a true square on the test surface and in contact with the bottom of the specimen. Rotate the specimen, keeping contact with the square, until the maximum gap between the square and the top of the specimen is found. Determine the width of the gap using a feeler gage. The perpendicularity tolerance is met if the gap, Δ , divided by the specimen length, L , is less than 1 part in 230, that is,

$$\frac{\Delta}{L} \leq \frac{1}{230} = 0.0043 \quad (2)$$

where:

L = length, and

Δ = length difference.

Repeat for other end of specimen unless the specimen ends were checked for parallelism in 9.2.2.

9.4 The measurements described in 9.1.2, 9.2.1, and 9.2.2 are taken with a mechanical dial gage. An optical or electronic device with an equivalent or better readout sensitivity and accuracy may be used in place of the dial gage, if desired.

9.5 The measurements taken on the ends of the specimen in 9.2.1 and 9.2.2 may also be made with the specimen clamped upright in a V-block. The measuring tip of the dial gage would then contact the upper end surface of the specimen. Either the V-block and specimen may be moved as a unit under a stationary dial gage, or the dial gage may be moved while the V-block and specimen remain stationary. The readings to be taken with such a vertical configuration are the same as those specified for the horizontal configuration in 9.2.1.

9.6 If the specimen does not pass the perpendicularity or flatness (or both) conformance criteria then the specimen should be evaluated to determine if a best effort was achieved or not for the rock type involved. Based upon the evaluation and professional judgment a determination is made whether the specimen should be discarded, tested as is, capped with a capping compound, or start over with at Section 8.4 (See Note 4).

10. Procedure for Determining Specimen Dimensional and Moisture Properties

10.1 General—The dimensions determined in this section yield three significant digits; therefore, calculated values of such items as stress, strain, density or unit weight can have only three significant digits. However, there might be cases requiring four significant digits. In that case, the dimension measurements would require four significant digits.

10.2 Diameter and cross sectional area—Determine the diameter of the test specimen perpendicular to the core axis, using a machinist caliper or similar device, to the nearest 0.01 in. (0.25 mm) by averaging two diameters measured at right angles to each other at about mid-height of the specimen.

10.3 Cross Sectional Area—Use the average diameter for calculating the circular cross-sectional area perpendicular to the core axis, to three significant digits.

10.4 Length—Determine the length of the test specimen using a machinist caliper, or similar device, to the nearest 0.01 in. (0.25 mm) at the centers of the end faces.

⁶ Podnieks, E. R., Chamberlain, P. G., and Thill, R. E., "Environmental Effects on Rock Properties," *Basic and Applied Rock Mechanics, Proceedings of Tenth Symposium on Rock Mechanics*, AIME, 1972, pp. 215–241.

10.5 *Volume*—Determine the volume of the specimen to three significant digits using the area and length values from Sections 10.3 and 10.4.

10.6 Record the moisture condition of the sample at the time of the receipt and at the completion of the specimen preparation procedure. At the minimum the moisture condition shall be reported qualitatively as as-received, saturated, laboratory air-dry, or oven dry. However, it is recommended that the moisture conditions be more precisely determined when possible and reported as either water content or degree of saturation using Test Method D2216.

11. Report: Test Data Sheet(s)/Form(s)

11.1 Report the following along with other data advised for the particular test method in 2.1 in which the specimen will be used:

11.1.1 Source of test specimen, including project name and location. Often the location is specified in terms of the drill hole number and depth of specimen from collar of hole.

11.1.2 Include curatorial record number if any, and curatorial history if pertinent to the specimen preparation.

11.1.3 Record the test method and testing apparatus for which the specimen preparations were intended.

11.1.4 Date, which practice method was performed, and personnel involved.

11.1.5 Physical description of test specimen including: rock type such as sandstone, limestone, granite, and alike.; location and orientation of inherent rock structural features and any discontinuities; and large inclusions or inhomogeneities, if any. A sketch of the test specimen is recommended for other than homogeneous rock types.

11.1.6 Specimen diameter and length, mass, L/D, volume, bulk specific gravity and conformance or best effort conformance with dimensional requirements (7.2, 7.3, and 7.4). When a specimen does not meet the requirements of this standard and 9.6 was followed and it is critical that it be tested or there are no other options (such as there are no other specimens), then allow the test specimen to be tested, but it shall be clearly noted in the report that it did not pass tolerances and by how much.

11.1.7 Data to substantiate the following three tolerances and which tolerances are applicable:

11.1.7.1 Straightness of the elements by either Procedure S1 (9.1.1) or Procedure S2 (9.1.2).

11.1.7.2 Flatness and parallelism by either Procedure FP1 (9.2.1) or Procedure FP2 (9.2.2).

11.1.7.3 Perpendicularity by either Procedure P1 (9.3.1) or Procedure P2 (9.3.2).

11.1.8 The moisture condition of the sample at the time it was received and after the specimen preparation is completed, such as as-received, saturated, laboratory air-dry, or oven dry. It is recommended that the moisture conditions be more precisely determined when possible and reported as either water content or degree of saturation.

11.1.9 List the equipment used to prepare the specimens and for conformance measurements.

11.1.10 Provide the raw data in the report if requested by the client.

12. Keywords

12.1 conformance; core; diameter; dimensional tolerances; dimensions; length; rock; specimen preparation (for testing); specimen shape; specimen size

SUMMARY OF CHANGES

Committee D18 has identified the location of selected changes to this standard since the last issue (D4543 – 08) that may impact the use of this standard. (June 1, 2019)

(1) Added a **Note 2** to explain some of the limitations on the method to take into account some of the testing labs having different load frames and preparing specimens from weak rock types.

(2) Units caveat was updated.

(3) New ASTM caveat added. This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.

(4) Reference “Needless Stringency in Sample Preparation Standards for Laboratory Testing of Weak Rocks” was added.

(5) **Note 3** was added about discussions on moisture content are common in many rock testing standards but professional judgement will be needed to both handle and report this issue.

(6) Subsection 5.1 editorially corrected. (0.0013 mm) to (0.013 mm).

(7) Subsection 5.13 was added for scales and the corresponding reference D4753 Guide for Evaluating, Selecting, and Specifying Balances and Standard Masses for Use in Soil, Rock, and Construction Materials Testing.

(8) **Note 5** was added about the equipment shown or discussed was not unique or required and other means and methods were okay to obtain the same results.

(9) Subsection 6.3 on Block samples being marked for overburden was added.

(10) **Note 7** was added to discuss the L/D ratio and the importance of size with respect to any shear planes.

(11) **Note 8** was added about pre-planning before starting the specimen preparation.

(12) **Fig. 2** was out of place and was moved.

(13) Old **Fig. 3** was changed to **Fig. 4** and updated and corrected as needed.

(14) **Note 12** was added for greater clarity about the displacement gage.

(15) Subsection 9.2.2 was reworded and expanded to provide more clarity.

(16) Added mass of specimen to Section 10.

(17) Corrected title for Section 11.

(18) Section 11.1.6 added wording on “best effort.”

(19) Section 11.1.8 added on including raw data if requested by client.

(20) Other minor revisions for clarity and alignment with current D18 or ASTM requirements.

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